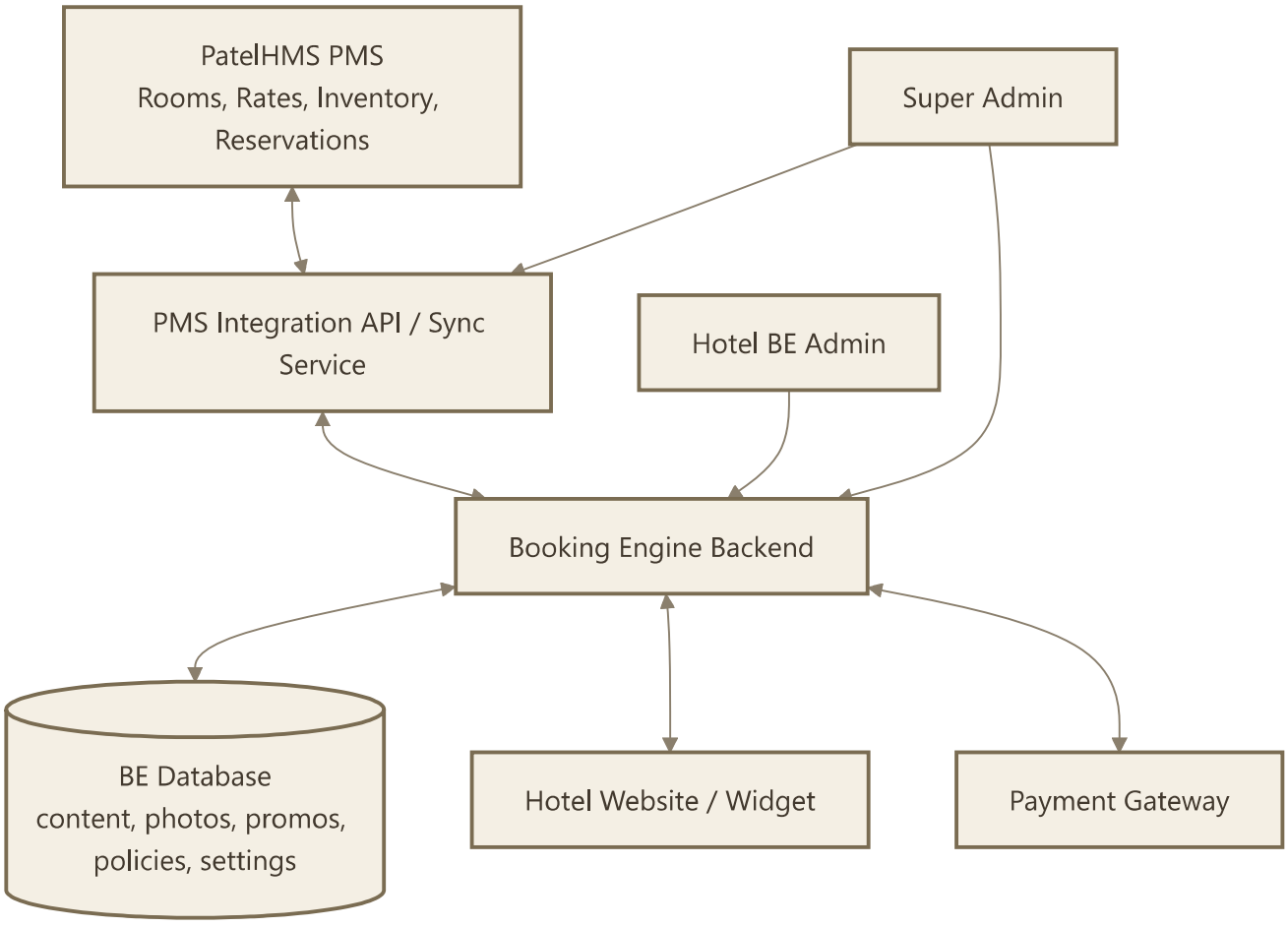
PatelHMS Booking Engine — Developer Build Spec

1. Core principle

Layer	Role
PMS	Source of truth: room types, inventory, rate plans, rates, availability, reservations, guests
Booking Engine (BE)	Customer-facing sales channel. Reads PMS data, writes reservations back to PMS
BE Admin (Hotel)	Controls presentation: photos, descriptions, which rates show, promos, policies, appearance
Super Admin (PatelHMS)	Activation, configuration, template, integrations, sync monitoring, logs

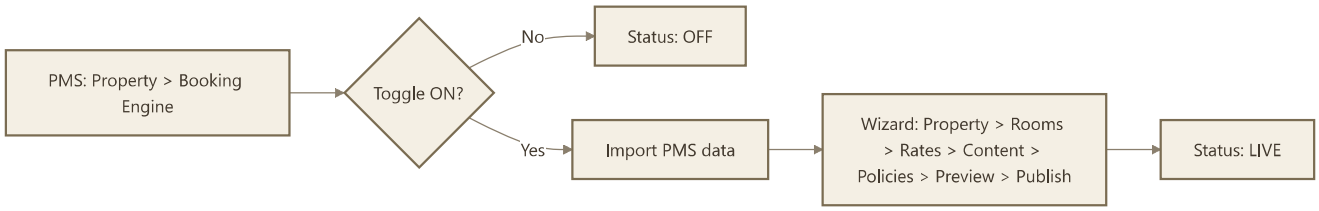
Rule: never duplicate rooms, rates or inventory inside the BE. BE stores only presentation and settings data, keyed by PMS IDs.

2. System architecture



3. Hotelier setup flow

3. Hotelier setup flow



Steps

1. **Enable** toggle in PMS creates a `booking_engine_property` record with status `SETUP_REQUIRED`.
2. **Auto-import from PMS** (read-only, no re-entry): property info (name, address, phone, email, check-in/out time, currency), room types (name, occupancy, bed type, amenities, inventory count), rate plans (base rates, occupancy pricing, restrictions), current availability.
3. **Wizard** (7 steps): Property, Rooms, Rates, Content, Policies, Preview, Publish. Each step is saved as draft; wizard can be resumed.
4. **Preview** (Desktop/Tablet/Mobile) must let the hotelier run the full test path: Home > Search > Room > Rate > Guest Details > Payment > Confirmation (test mode, no real reservation).
5. **Publish** sets status `LIVE`, generates URL (e.g. `book.hotelname.com`), logs the event.

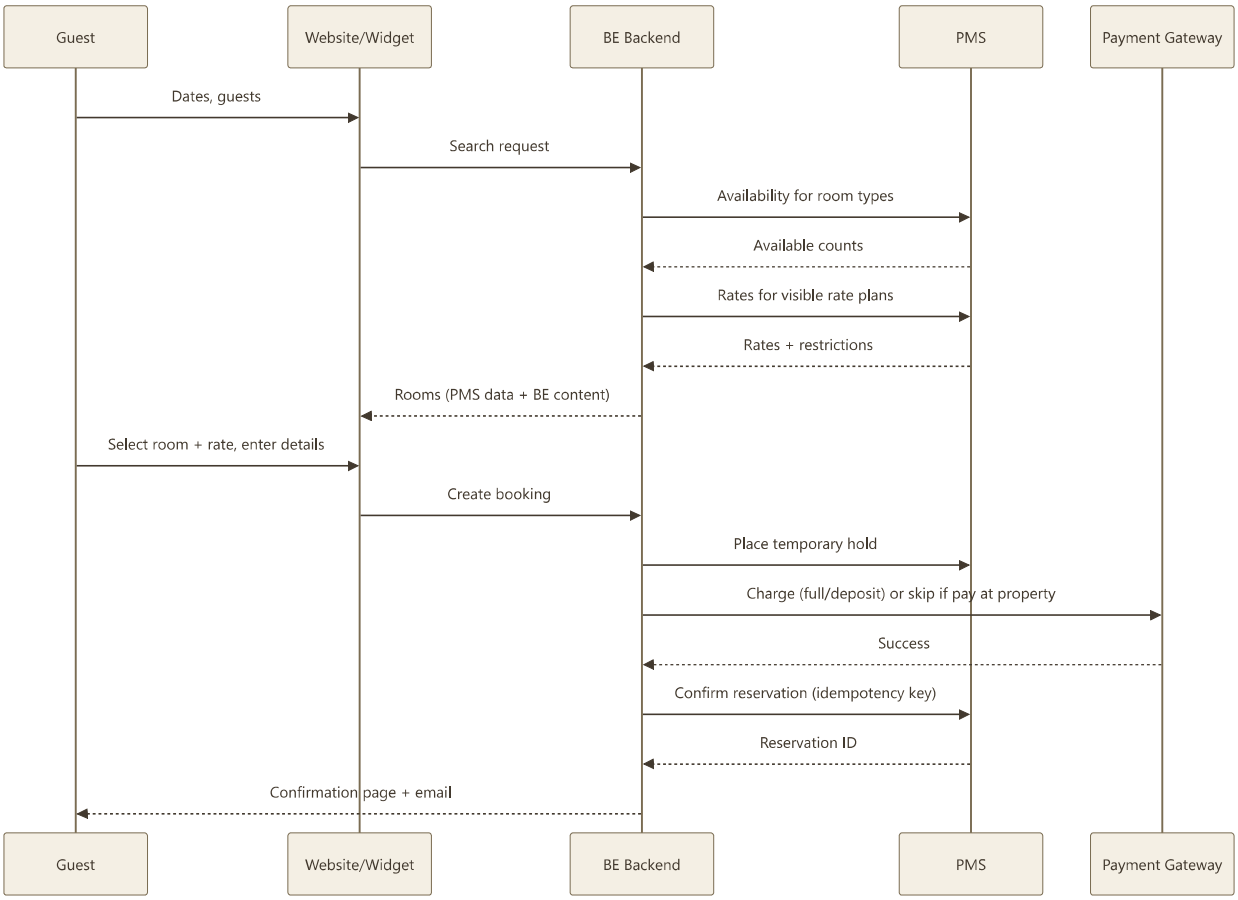
Status values: `OFF`, `SETUP_REQUIRED`, `READY`, `LIVE`, `DISABLED` (by super admin).

4. Hotel Admin modules

Module	What it does	Data owner
Dashboard	Status, bookings, revenue, sync health	BE
Website Content	Hotel name, short/about text, amenities, hotel photos (upload, delete, reorder, set primary)	BE
Rooms	Per PMS room type: display name, description, photos, amenities, display order, show/hide	BE (presentation) over PMS room type
Rates & Inventory	Per PMS rate plan: "Show on Booking Engine" yes/no. Inventory shown read-only	PMS (BE stores only the visibility flag)
Promotions	Promo codes (%, fixed), date range, applicable room types; offers like "Stay 3 Get 1 Free"	BE
Policies	Cancellation, child, pet, check-in, payment, T&C; shown during booking	BE
Booking Settings	Required/optional guest fields; min/max stay, advance window, cut-off, max rooms per booking	BE
Payments	Full / deposit / pay at property; gateway connection	BE + gateway
Appearance	Logo, favicon, primary/secondary colour, hero image, headline, about section. Template-controlled, not a website	BE

	builder	
Preview / Publish	See section 3	BE

5. Guest booking flow



Rules

- Availability and price are **always fetched live from PMS** (short cache only, e.g. under 60 seconds).
- Re-check availability and price immediately before payment.
- Use a **temporary hold** (e.g. 10 minutes) on the room while the guest pays, released on timeout or failure.
- Reservation creation must be **idempotent** (idempotency key per booking attempt) to prevent duplicates on retries.
- If payment succeeds but PMS confirmation fails: auto-retry, then raise an alert and mark `NEEDS_ATTENTION` in Super Admin (never silently lose a paid booking).
- Reservations carry source = `BOOKING_ENGINE` and promo code if used.

6. Two-way sync (the loop)

Event	Flow
Front desk books a room	PMS inventory decreases, BE availability reflects it (no manual update)

	manual update)
Guest books on website	BE creates PMS reservation, inventory decreases
Cancellation/modification in PMS	Inventory released, BE availability updates
Rate changed in PMS Rate Calendar / Bulk Rate Update	BE shows new rate on next search
Room type or rate plan added/removed in PMS	Appears/disappears in BE Admin (new items default to hidden until the hotelier enables them)

Implementation options: (a) BE calls PMS API live on every search (simplest, recommended for v1); (b) PMS pushes webhooks and BE keeps a short-lived cache. Start with (a), add (b) if performance requires it.

7. Data model (BE side only)

Use PMS IDs as foreign references; do not copy PMS operational fields.

- `be_property` (pms_property_id, status, url, published_at, template_id)
- `be_property_content` (property_id, short_description, about, amenities, hero_image, headline)
- `be_room_content` (pms_room_type_id, display_name, description, amenities, display_order, visible)
- `be_media` (owner_type: property/room, owner_id, file_url, sort_order, is_primary)
- `be_rate_visibility` (pms_rate_plan_id, show_on_be)
- `be_promotion` (code, type, value, start, end, applicable_room_type_ids)
- `be_policy` (property_id, type, content, cancellation_rules)
- `be_booking_settings` (required_fields, min_stay, max_stay, advance_window, cutoff, max_rooms)
- `be_payment_settings` (mode: full/deposit/pay_at_property, deposit_value, gateway_status)
- `be_appearance` (logo, favicon, primary_color, secondary_color)
- `be_booking_attempt` (id, idempotency_key, status, pms_reservation_id, payment_ref, hold_expires_at)
- `be_sync_log` (property_id, event_type, status, payload_ref, timestamp)

8. Integration API (PMS contract)

Developer should confirm or build these PMS endpoints:

- `GET property`, `GET room-types`, `GET rate-plans`
- `GET availability?from&to&adults&children` (counts per room type)
- `GET rates?from&to&room_type&rate_plan` (including restrictions)
- `POST holds` / `DELETE holds/{id}`
- `POST reservations` (with idempotency key, source, guest, payment info)
- `GET/PUT reservations/{id}` (status, cancel)
- Webhooks (phase 2): inventory, rate, reservation changes

9. Super Admin (PatelHMS internal)

- Property list with status (Active / Setup Required / Disabled)

- Property list with status: 🟢 Active, 🟡 Setup required, 🔴 Disabled
- Per property: Configuration, Website/template, Inventory view, Bookings, Payments, Logs
- Health indicators: BE status, PMS connection, Payment connection, Last sync time
- Controls: enable/disable BE, assign template, set URL/domain, retry failed bookings
- **Sync / Activity Log** (searchable), e.g. "Rate updated", "Inventory updated", "Reservation #10234 created", "Room photo updated", "BE published"

10. Roles and permissions

- **Hotelier:** edit content/photos, rate visibility, promos, policies, booking settings, appearance, preview, publish.
- **Hotelier cannot:** create rooms, rates or inventory in BE (done in PMS only).
- **Super Admin:** everything above plus activation, integrations, logs, support actions.

11. Edge cases to handle

- Double booking (front desk and guest simultaneously): solved by holds + final availability check.
- Price changes mid-checkout: re-validate, show the guest the new price before charging.
- Hold expiry during payment.
- Payment success but reservation failure (see section 5).
- Restrictions: min stay, closed to arrival/departure, stop-sell from PMS must be honoured.
- Room type with zero visible rates: hide the room from BE.
- Out-of-order / maintenance rooms excluded from availability.
- Timezone and currency from PMS property settings.
- Promo stacking, validity dates, and room applicability.

12. Build phases

Phase 1 — Foundation: PMS integration API, BE database, Super Admin enable/disable, PMS data import. **Phase 2 — Core booking:** Search > Room > Rate > Guest details > Reservation in PMS (pay at property first), holds, idempotency, confirmation email. **Phase 3 — Payments:** Gateway integration, full/deposit modes, failure handling. **Phase 4 — Hotel Admin:** Content, photos, rooms presentation, rate visibility, policies, booking settings, appearance. **Phase 5 — Setup wizard, Preview, Publish**, URL/domain handling. **Phase 6 — Promotions**, dashboard, Super Admin monitoring and sync logs. **Later:** OTA/channel manager, dual pricing (direct vs OTA), webhooks for push sync.

13. Acceptance criteria (summary)

1. Turning BE ON imports PMS data with zero re-entry.
2. Hotelier edits photos/content and sees changes in Preview before Publish.
3. Hiding a rate plan removes it from the website immediately.
4. A website booking appears in PMS as a reservation and reduces inventory.
5. A front-desk booking reduces BE availability with no manual step.
6. A PMS rate change shows on the website on the next search.

7. No duplicate reservations on retry; no paid booking lost without an alert.
8. Super Admin sees status, connection health, last sync and a full activity log per property.